

Exploring Amyloid- β Interactions with Multicomponent Neuronal Membranes

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ABSTRACT

The aggregation of amyloid- β (A β) peptides and their interactions with neuronal membranes are central to the pathology of Alzheimer's disease [1,2]. Although the two predominant isoforms, A β_{40} and A β_{42} , are known to associate with lipid bilayers, their molecular behaviour in compositionally complex neuronal membranes remains incompletely understood. Here, we use all-atom molecular dynamics simulations to investigate the interactions of initially disordered A β_{40} and A β_{42} monomers with two membrane models of increasing complexity: a ternary POPC/SM/CHOL (2:1:1) bilayer and a five-component neuronal membrane containing POPC, DPPC, SM, CHOL, and GM1. We characterize peptide conformations, residue-specific peptide-lipid interactions, lipid-contact patterns, lipid preferences, and selected structural properties of the membranes. By systematically comparing the two peptide isoforms across these membrane environments, this study explores the relationship between membrane composition, peptide conformational behaviour, and peptide-lipid association. This work provides a molecular-level framework for examining A β -membrane interactions in biologically relevant lipid environments and contributes to a broader understanding of membrane-associated processes potentially involved in the early stages of Alzheimer's disease.

REFERENCES

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